

SAMYAK MOON

Bengaluru, Karnataka

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PROFESSIONAL SUMMARY

Backend Software Engineer with 2.8+ years of experience designing and building scalable, secure fintech systems using **Java** and **Spring Boot**. Experienced in **microservices**, **distributed systems**, and **payment/card platforms** with a strong focus on performance, reliability, and system design.

EXPERIENCE

Software Engineer, PayU Digital Labs

April 2025 - Present

Bengaluru, Karnataka

- Designed and owned the **Card Order Workflow** end-to-end, implementing a **maker-checker approval pattern** and **Quartz-based async job** scheduling for **Visa** and **RuPay** card generation across multiple clients; optimised for **parallel processing**, reducing card generation time for **10K+ cards** by **85%**.
- Build **SecureAuthPro**, a **multi-tenant Authentication Authorization** service using **Spring Boot**, **Redis**, and **RSA-256 JWT**; enforced strict tenant isolation, **RBAC**, and **OTP-based login** across 10+ enterprise clients, meeting enterprise and regulatory security standards.
- Designed **Wallet-to-Account Transfer (W2A)** API with **idempotency guarantees**, **concurrency control**, and **retry mechanisms**; implemented a **Quartz-based async refund job** to handle failed transactions, ensuring data consistency and financial reconciliation at scale.
- Enhanced **Authorization Settlement** flows for **Visa** and **RuPay** network transactions across **e-commerce**, **POS**, and **ATM channels**; reduced processing latency and integrated an **External authentication mechanism** enabling clients to Approve/Reject network transactions in real time.
- Integrated **Atalla Payment HSM** for **PIN management**, **key lifecycle operations**, and **PCI-DSS compliant cryptographic operations**; achieved **60%** faster transaction processing vs. a general-purpose HSM, reducing integration risk.
- Implemented **Two-Factor Authentication** (MPIN + OTP) for financial transactions using **Redis TTL-based OTP management**; designed flow to align with **RBI** and **PCI-DSS security expectations**, strengthening **fraud prevention** and access control with low-latency processing.

PROJECTS

Splitmoney – Real-Time Expense Splitting WebApp

[Github](#)

- Integrated Card UPI Payments Enabled real-time debt resolution via Hyperswitch with **webhook-driven async processing**, **idempotent balance updates**, and event-sourced reconciliation across microservices.
- gRPC Inter-Service Communication** Replaced REST with **gRPC + Protobuf** for internal expense → payment service calls, achieving lower latency and strict contract enforcement.
- Multi-Strategy Expense Splitting** Built equal, exact-amount, and percentage-based splits with per-group net-balance tracking and idempotent expense creation via SHA-256 request hashing.
- Reliable **Event-Driven Notifications** Guaranteed at-least-once delivery using **Transactional Outbox**, **RabbitMQ dead-letter queues**, and **Redis-based deduplication** — eliminating dual-write inconsistencies.

TECHNICAL SKILLS

Programming	Java, Python, SQL
Backend & Distributed Systems	Spring Boot, REST APIs, Microservices, Multi-Tenancy
Databases & Storage	MySQL, PostgreSQL, Redis, Elasticsearch
Cloud & DevOps	AWS, Jenkins, CI/CD
Messaging & Streaming	Kafka, RabbitMQ
Observability & Monitoring	ELK Stack(Elasticsearch, Logstash, Kibana), Grafana
Technologies	Cursor AI (Advanced), LLM-powered coding, Generative AI tools.

EDUCATION

M.Tech in Computer Science

National Institute of Technology Surathkal, Karnataka

2021 - 2023

B.Tech in Computer Science

Government College of Engineering Amravati

2016 - 2020